Patient Calibrated Dynamic Graph Fusion for Cross-Patient EEG Seizure Detection

ESMAIL IBRAHIM MOHAMMED AL-NAKEEB
ID: 2598555
Bahçeşehir University
Master of Science in Artificial Intelligence
alnakeebesmail@gmail.com

Abstract

Cross-patient EEG seizure detection is challenging because seizure patterns and normal brain activity can vary strongly between patients. This project proposes a patient-calibrated dynamic graph fusion framework for seizure detection using scalp EEG recordings from the CHB-MIT dataset. Each EEG window is converted into a dynamic brain connectivity graph, where EEG channels are represented as nodes and correlation-based functional relationships are represented as edges. In addition to standard dynamic graph learning, patient-specific normal baseline graphs are estimated from reserved non-seizure calibration windows and used to construct patient-calibrated graph representations. Several methods were compared, including logistic regression, random forest, SVM, MLP, static GCN, dynamic GCN, dynamic GAT, patient-calibrated GAT, hybrid DPC GAT, and a validation-tuned graph ensemble. The final ensemble achieved the best performance on the primary patient level test split with an F1 score of 0.8061, ROC AUC of 0.9836, recall sensitivity of 0.9813, and false positive rate of 0.1858. Compared with the strongest individual baseline, Dynamic GCN, the ensemble improved F1 score and reduced false positives while maintaining very high seizure sensitivity. Bootstrap analysis supported this improvement trend on the primary test split, while repeated patient-split testing showed that the performance gain is patient-dependent. These results suggest that patient-calibrated graph information can improve cross-patient EEG seizure detection when combined with validation-tuned dynamic graph fusion, although patient variability remains a major limitation.

1. Introduction

Epileptic seizure detection is an important biomedical signal processing problem because seizures can appear suddenly and may require fast clinical attention. Electroencephalography, usually referred to as EEG, is one of the most common tools for monitoring brain activity because it records electrical signals from multiple scalp electrodes over time. In seizure detection, the goal is to distinguish seizure activity from normal background EEG. Although this may seem like a binary classification problem, it is difficult because EEG signals are noisy, nonstationary, and highly variable between patients.

A major challenge in EEG seizure detection is cross-patient generalization. A model can perform well when it is trained and tested on windows from the same patient, but this does not necessarily mean that it will work on unseen patients. Randomly splitting EEG windows can create data leakage because nearby windows from the same recording share similar temporal and patient-specific patterns. For this reason, this project uses a patient-level split where the training, validation, and test sets contain different patients. This makes the evaluation harder, but it better reflects the real objective of detecting seizures in patients that were not used during model training.

Traditional machine learning methods usually depend on handcrafted EEG features extracted from time and frequency domains. Deep learning methods can learn more flexible patterns, but many standard approaches treat EEG windows as flat feature vectors or temporal sequences. This may ignore the spatial relationships between EEG channels. Since each EEG channel corresponds to a scalp location, the relationships between channels can be represented as a graph. In this project, each EEG window is converted into a dynamic brain connectivity graph, where nodes represent EEG channels and edges represent correlation-based functional relationships between channels.

The main idea of this project is that seizure-related brain connectivity should not only be modeled globally but also interpreted relative to the patient's normal EEG behavior. Therefore, patient-specific baseline graphs are estimated from reserved non-seizure calibration windows. These baseline graphs are used to create patient-calibrated graph representations. The project then compares several approaches, including traditional classifiers, MLP, static GCN, dynamic GCN, dynamic GAT, patient-calibrated GAT, hybrid DPC GAT, and a validation-tuned graph ensemble.

The main contribution of this work is a patient-calibrated dynamic graph fusion framework for cross-patient EEG seizure detection. The final ensemble combines standard dynamic graph learning with patient-calibrated graph information using validation-selected weights and thresholding. The objective is not only to improve classification accuracy, but also to reduce false alarms while preserving high seizure sensitivity. The experiments show that the proposed ensemble achieved the best result on the primary patient level test split, improving F1 score and reducing false positives compared with the strongest individual graph baseline. Additional bootstrap and repeated patient split tests were included to evaluate the stability and limitations of the result.

2. Related Work

EEG seizure detection has been studied using both traditional machine learning and deep learning methods [5, 8]. Early machine learning approaches usually depend on handcrafted features extracted from EEG signals, such as statistical features, spectral power, signal energy, and frequency band information. These features are then passed to classifiers such as support vector machines, random forests, logistic regression, or other conventional models. These methods are useful because they are relatively interpretable and computationally efficient, but they can struggle when seizure patterns vary strongly between patients.

Public EEG datasets have made seizure detection research more reproducible. In this project, the CHB-MIT scalp EEG database is used because it contains pediatric EEG recordings with seizure annotations and is widely used in seizure detection studies [1, 2, 7]. The dataset contains EDF recordings and seizure start and end times, which makes it suitable for supervised window-based seizure detection. However, EEG data from this dataset is still challenging because different patients can have different seizure characteristics, channel behavior, and normal background activity.

Deep learning methods have also been applied to EEG seizure detection because they can learn more complex patterns than traditional feature-based classifiers [5]. Neural networks can model nonlinear relationships in EEG windows, and models such as MLPs, CNNs, recurrent networks, and attention-based networks have been used in biomedical signal classification. However, many deep learning approaches treat EEG either as a flat vector or as a temporal sequence. This can ignore the spatial and functional relationships between EEG electrodes.

Graph neural networks provide a natural way to model EEG because EEG channels can be treated as graph nodes. The edges can represent anatomical relationships, physical electrode distances, or functional connectivity between channels. Graph convolutional networks introduced the idea of learning node representations by aggregating information from graph neighbors [4]. Graph Attention Networks extended this idea by allowing the model to learn the importance of neighboring nodes through attention weights [9]. These methods are suitable for EEG because seizure activity may appear as abnormal patterns of connectivity across multiple channels rather than only as changes in a single channel.

Recent EEG graph learning studies show that graph based models can be useful for seizure detection, especially when the connectivity between electrodes is informative. Graph-based EEG seizure prediction has also been studied using scalp EEG, which supports the idea of representing EEG channels as graph nodes [3]. Other work has also explored hybrid graph-based seizure detection and cross-patient evaluation on the CHB-MIT dataset [6]. These studies support the motivation for using graph-based EEG representations instead of only flat feature vectors.

However, cross-patient seizure detection remains difficult because the normal EEG baseline and seizure expression differ across patients. A model trained on one group of patients may not generalize well to unseen patients if it only learns global patterns. This motivates patient-adaptive approaches, where each patient is interpreted relative to their own normal EEG behavior.

The method in this project follows this direction. Instead of using only a fixed graph or only a dynamic graph, the project estimates patient-specific normal baseline graphs from reserved non-seizure calibration windows. These baseline graphs are used to construct patient-calibrated graph representations. The final system then combines standard dynamic graph learning with patient-calibrated graph attention through a validation-tuned ensemble. This makes the project different from a simple classifier comparison because it studies how patient-specific graph calibration affects cross-patient seizure detection.

3. Data Description

This project uses the CHB-MIT scalp EEG database, which is a public EEG dataset available through PhysioNet [1, 7]. The dataset contains scalp EEG recordings collected from pediatric patients with epileptic seizures. The recordings are stored as EDF files, and the dataset provides seizure start and end annotations for seizure containing recordings. These annotations make it possible to build a supervised seizure detection dataset by dividing the continuous EEG signals into labeled windows.

Table 1. Processed dataset summary.

Item	Value
Selected patients	8
Graph nodes	18
Sampling rate	128 Hz
Window length	4 seconds
Stride	2 seconds
Total windows	2712
Seizure windows	678
Non seizure windows	2034

For this project, a controlled subset of patients was used to keep the full experiment executable in Google Colab. The selected patients were chb01, chb02, chb03, chb05, chb06, chb07, chb08, and chb10. For each selected patient, two seizure containing EDF files were downloaded automatically from PhysioNet when available. The raw EDF files were not included in the final code package because they are large. Instead, the notebook downloads the selected subset directly from the public dataset source.

A common set of bipolar EEG channels was selected so that all EEG windows could be represented using the same graph structure. The selected channels were FP1 F7, F7 T7, T7 P7, P7 O1, FP1 F3, F3 C3, C3 P3, P3 O1, FP2 F4, F4 C4, C4 P4, P4 O2, FP2 F8, F8 T8, T8 P8, P8 O2, FZ CZ, and CZ PZ. Each channel was treated as one graph node, so every EEG graph had 18 nodes.

The EEG recordings were resampled to 128 Hz and filtered using a bandpass filter from 0.5 Hz to 45 Hz. Each recording was divided into 4 second windows with a 2 second stride. A window was labeled as seizure if it overlapped with any annotated seizure interval. Non seizure windows were sampled from the remaining parts of the recording to reduce class imbalance while keeping enough normal EEG examples for training and evaluation.

The final processed dataset contained 2712 EEG windows. Of these, 678 windows were seizure windows and 2034 windows were non seizure windows. Each window was represented by node level EEG features and a dynamic graph adjacency matrix. The node features included time domain and frequency domain information, while the adjacency matrix represented correlation based connectivity between EEG channels.

The dataset was split by patient rather than by random EEG windows. This is important because random window splitting can cause data leakage when windows from the same patient appear in both training and testing. In the primary experiment, the training patients were chb05, chb06, chb08, and chb10. The validation patient was chb03, and the test patients were chb07, chb01, and chb02. A small number of normal windows from validation and test patients were reserved only for patient calibration. These calibration windows were not used when calculating validation or test performance.

Table 2. Primary patient level split used in the main experiment.

Split	Patients
Training	chb05, chb06, chb08, chb10
Validation	chb03
Testing	chb07, chb01, chb02

This patient level split makes the experiment more difficult than a random split, but it gives a more realistic evaluation of cross patient seizure detection. The purpose is to test whether the model can generalize to patients that were not used during training.

4. Methods

The proposed pipeline converts raw EEG recordings into window level graph representations and then trains several baseline and graph based models for seizure detection. The complete workflow consists of EEG preprocessing, window labeling, node feature extraction, dynamic graph construction, patient baseline calibration, graph model training, validation tuned fusion, and final evaluation.

4.1. Overall Workflow

The complete methodology can be summarized as follows:

1. Download selected seizure containing EDF files from the CHB-MIT dataset.
2. Read seizure start and end annotations from the patient summary files.
3. Select common bipolar EEG channels across recordings.
4. Resample the EEG signal to 128 Hz and apply bandpass filtering between 0.5 Hz and 45 Hz.
5. Divide each EEG recording into 4 second windows with a 2 second stride.
6. Label each window as seizure or non seizure based on overlap with annotated seizure intervals.
7. Extract time domain and frequency domain node features from each EEG channel.
8. Construct a dynamic connectivity graph for each EEG window using channel correlation.
9. Estimate patient normal baseline graphs from reserved non seizure calibration windows.
10. Train traditional baselines, MLP, GCN, GAT, patient calibrated GAT, hybrid DPC GAT, and the final validation tuned graph ensemble.
11. Evaluate all methods on unseen test patients using accuracy, precision, recall sensitivity, specificity, F1 score, ROC AUC, false positive rate, and false negative rate.

This design was chosen to avoid a simple random window classification setup. The most important methodolog-

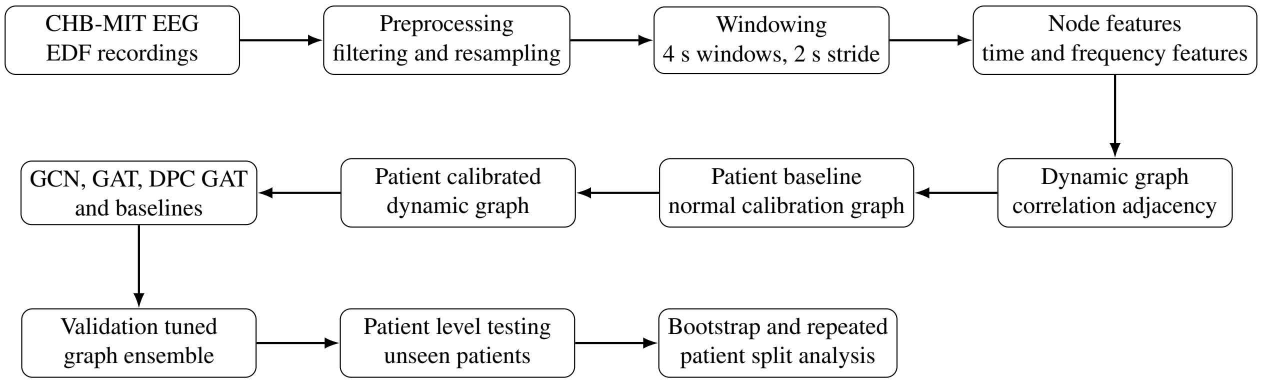

Figure 1. Main workflow of the proposed patient calibrated dynamic graph fusion pipeline.

ical decision is the patient level split, where training, validation, and testing are performed on different patients. This makes the task harder but gives a more realistic evaluation of cross patient seizure detection.

4.2. EEG Window Representation

Each EEG recording was converted into fixed length windows. A window length of 4 seconds was used because it is long enough to capture short seizure related signal changes while still producing a sufficient number of training examples. A stride of 2 seconds was used to create overlapping windows.

For each EEG window, every EEG channel was treated as one node in a graph. Therefore, each window was represented as a graph with 18 nodes. A feature vector was extracted from each node. The extracted node features included mean, standard deviation, signal energy, line length, zero crossing rate, and frequency band power features. These features give each node information about both time domain and frequency domain behavior.

Let $X_i \in \mathbb{R}^{N\times F}$ represent the node feature matrix for EEG window i, where $N = 18$ is the number of EEG channels and F is the number of node features. Each row of X_i corresponds to one EEG channel.

4.3. Dynamic Graph Construction

For each EEG window, a dynamic graph adjacency matrix was constructed using absolute Pearson correlation between EEG channels. This means that every window has its own graph structure instead of using one fixed graph for all examples. The goal is to capture functional connectivity changes between EEG channels during seizure and non seizure states.

For a given window, the correlation between two EEG channels was computed and converted to an absolute value. To reduce graph noise, only the strongest connections were retained using a top k edge selection strategy. In this project, the top 5 strongest connections for each node were used. The graph was then symmetrized and normalized before being passed to the graph neural network models.

Let $A_i \in \mathbb{R}^{N\times N}$ represent the dynamic adjacency matrix for EEG window i. The normalized adjacency matrix was computed as:

$$\hat{A}_i = D_i^{-\frac{1}{2}} A_i D_i^{-\frac{1}{2}}, \tag{1}$$

where D_i is the degree matrix of A_i.

4.4. Patient Baseline Graph Calibration

The main contribution of the project is the use of patient specific graph calibration. Cross patient seizure detection is difficult because the normal EEG connectivity pattern of one patient may be different from another patient. Therefore, instead of treating all patient graphs in the same global way, this project estimates a normal baseline graph for each patient.

For training patients, the baseline graph was computed from their non seizure training windows. For validation and test patients, a small number of reserved non seizure calibration windows was used. These calibration windows were not included in validation or test scoring. This prevents the model from being evaluated on windows that were also used for calibration.

For patient p, the baseline graph B_p was computed as the average graph over the available normal windows:

$$B_p = \frac{1}{M_p} \sum_{i=1}^{M_p} A_i, \tag{2}$$

where M_p is the number of available non seizure calibration windows for patient p. The difference between a current EEG graph and the patient baseline graph was used to measure how much the current brain connectivity deviates from the patient's normal state.

4.5. Patient Calibrated Dynamic Graphs

After estimating the patient baseline graph, each dynamic graph was recalibrated using its deviation from the patient normal baseline. The purpose of this step is to make abnormal graph changes more visible to the model.

For window i belonging to patient p, the graph deviation was computed as:

$$\Delta_i = |A_i - B_p|. \tag{3}$$

The patient calibrated adjacency matrix was then defined as:

$$A_i^{cal} = A_i \odot (1 + \alpha\Delta_i), \tag{4}$$

where $\odot$ denotes element wise multiplication and α controls the strength of patient calibration. Several calibration values were tested: 0.25, 0.50, 0.75, 1.00, 1.50, and 2.00.

Two patient calibrated model variants were tested. The first variant used the calibrated adjacency matrix only and was called Patient Calibrated GAT. The second variant also added a node level graph deviation feature to the original node features and was called Hybrid Deviation Patient Calibrated GAT, abbreviated as Hybrid DPC GAT. This hybrid design allows the model to see both the original EEG node features and how much each node deviates from the patient's normal graph behavior.

4.6. Graph Neural Network Models

Several graph based models were implemented. A Static GCN used one average training graph for all EEG windows. This baseline tests whether a fixed graph structure is enough. A Dynamic GCN used the window specific graph A_i, allowing each EEG window to have its own connectivity structure. A Dynamic GAT used graph attention to learn the importance of neighboring EEG channels.

The graph convolution operation follows the general form:

$$H^{(l+1)} = \sigma(\hat{A}H^{(l)}W^{(l)}), \tag{5}$$

where $H^{(l)}$ is the node representation at layer l, $W^{(l)}$ is a trainable weight matrix, $\hat{A}$ is the normalized adjacency matrix, and σ is a nonlinear activation function. The final graph representation was obtained by averaging node embeddings across all EEG channels, followed by a classifier that predicts seizure or non seizure.

The graph attention model follows the same graph classification idea, but it learns attention weights between connected nodes instead of applying the same aggregation rule to all neighbors. This allows the model to focus more strongly on EEG channel relationships that are useful for seizure detection.

4.7. Validation Tuned Graph Ensemble

The final method is a validation tuned graph ensemble. Instead of selecting only one graph model, the ensemble combines the predicted seizure probabilities from the strongest graph based models. The candidate models included Dynamic GCN, Dynamic GAT, the strongest Patient Calibrated GAT models, and the strongest Hybrid DPC GAT models.

For a set of candidate models, the final ensemble probability was calculated as:

$$P_{\text{ens}}(y = 1 \mid x) = \sum_{m=1}^{M} w_m P_m(y = 1 \mid x), \tag{6}$$

where $P_m(y = 1 \mid x)$ is the seizure probability predicted by model m, and w_m is the validation selected weight for that model. The final classification threshold was also selected using validation performance. The test set was used only once for final evaluation after the ensemble weights and threshold had been selected.

This design makes the final model more robust than relying on one individual graph model. It allows the system to combine the stability of Dynamic GCN and Dynamic GAT with the patient specific information learned by patient calibrated graph attention models.

4.8. Evaluation Strategy

The project used several levels of evaluation. First, all methods were compared on the primary patient level test split. Second, an ablation table was created to compare classical machine learning, flat neural network learning, static graph learning, dynamic graph learning, patient calibrated graph learning, and validation tuned graph fusion. Third, patient wise results were reported to show how the final method behaved on each unseen test patient. Fourth, bootstrap resampling was used to compare the final ensemble against the strongest individual graph baseline on the primary test split. Finally, repeated patient split testing was performed to check whether the improvement was stable across different patient train validation test assignments.

This evaluation strategy was used to avoid overclaiming based on only one metric or one split. The goal was not only to report the best result, but also to show where the proposed patient calibrated graph fusion method works well and where it remains limited.

5. Experiments and Results

This section presents the experimental setup, baseline comparisons, final model performance, ablation study, patient wise analysis, bootstrap testing, and repeated patient split robustness testing. The main goal of the experiments was to evaluate whether patient calibrated graph information can improve cross patient EEG seizure detection compared with traditional classifiers and individual graph neural network models.

5.1. Experimental Setup

All experiments were run in Google Colab using the processed CHB-MIT EEG graph dataset described earlier. The primary split used different patients for training, validation, and testing. Model selection, ensemble weights, and classification thresholds were selected using the validation set only. The test set was used only for final evaluation.

The evaluated methods included Logistic Regression, Random Forest, SVM with RBF kernel, MLP, Static GCN, Dynamic GCN, Dynamic GAT, Patient Calibrated GAT, Hybrid DPC GAT, and the final Validation Tuned Graph Ensemble. Traditional machine learning methods used flattened EEG node features, while graph models used EEG channels as graph nodes and functional connectivity as graph edges.

The evaluation metrics were accuracy, precision, recall sensitivity, specificity, F1 score, ROC AUC, false positive rate, and false negative rate. Recall sensitivity is important because missed seizure windows are clinically serious. False positive rate is also important because excessive false alarms reduce the usefulness of seizure detection systems.

5.2. Final Model Comparison

Table 3 shows the main comparison between the final ensemble and the strongest baseline methods. The Validation Tuned Graph Ensemble achieved the best F1 score on the primary patient level test split. It reached an F1 score of 0.8061, accuracy of 0.8628, ROC AUC of 0.9836, and recall sensitivity of 0.9813.

Compared with the strongest individual graph baseline, Dynamic GCN, the final ensemble improved the F1 score from 0.7903 to 0.8061. The false positive rate was reduced from 0.2088 to 0.1858. In terms of confusion matrix values, Dynamic GCN produced 109 false positives and 3 false negatives, while the final ensemble produced 97 false positives and 4 false negatives. This means that the ensemble reduced false alarms by 12 windows while maintaining very high seizure sensitivity.

5.3. Main Result Visualization

Figure 2 shows the F1 score comparison across the main methods. The final validation tuned graph ensemble achieved the highest F1 score. Figure 3 shows the false positive rate comparison. Although SVM achieved the lowest false positive rate, its recall was much lower than the graph based methods. The final ensemble gave a better balance between high seizure sensitivity and reduced false positives.

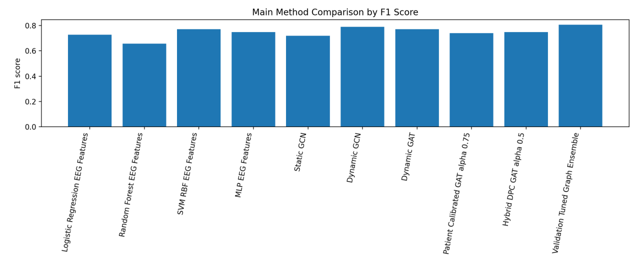

Figure 2. Main method comparison by F1 score on the primary test split.

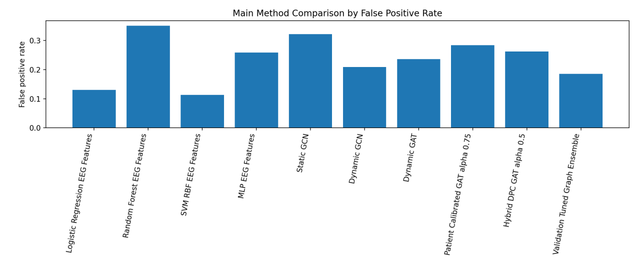

Figure 3. Main method comparison by false positive rate on the primary test split.

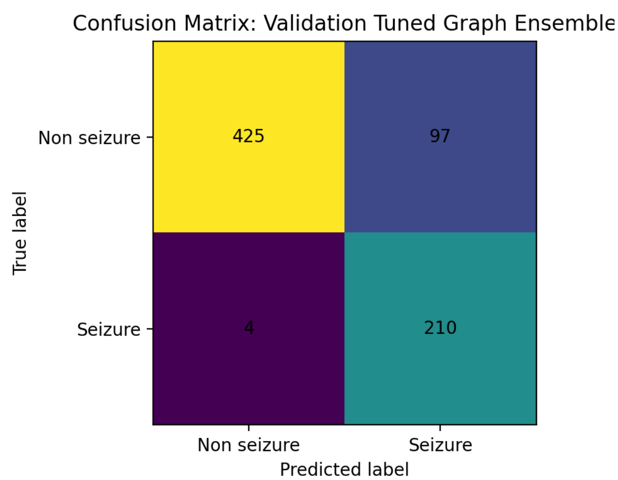

Figure 4. Confusion matrix of the final Validation Tuned Graph Ensemble.

5.4. Final Model Confusion Matrix and ROC Curve

The confusion matrix of the final ensemble is shown in Figure 4. The model correctly detected 210 seizure windows and missed only 4 seizure windows. It also correctly classified 425 non seizure windows but produced 97 false positive windows.

The ROC curve in Figure 5 shows that the final ensemble produced strong probability separation between seizure and non seizure windows, with ROC AUC of 0.9836.

Table 3. Final test comparison of the main methods on the primary patient level split.

Method	Accuracy	Precision	Recall	Specificity	F1	FPR
Validation Tuned Graph Ensemble	0.8628	0.6840	0.9813	0.8142	0.8061	0.1858
Dynamic GCN	0.8478	0.6594	0.9860	0.7912	0.7903	0.2088
SVM RBF EEG Features	0.8614	0.7435	0.7991	0.8870	0.7703	0.1130
Dynamic GAT	0.8288	0.6325	0.9860	0.7644	0.7701	0.2356
MLP EEG Features	0.8098	0.6076	0.9766	0.7414	0.7491	0.2586
Hybrid DPC GAT alpha 0.5	0.8084	0.6052	0.9813	0.7375	0.7487	0.2625
Patient Calibrated GAT alpha 0.75	0.7976	0.5900	0.9953	0.7165	0.7409	0.2835
Logistic Regression EEG Features	0.8356	0.7031	0.7523	0.8697	0.7269	0.1303
Static GCN	0.7717	0.5602	1.0000	0.6782	0.7181	0.3218
Random Forest EEG Features	0.7242	0.5146	0.9065	0.6494	0.6565	0.3506

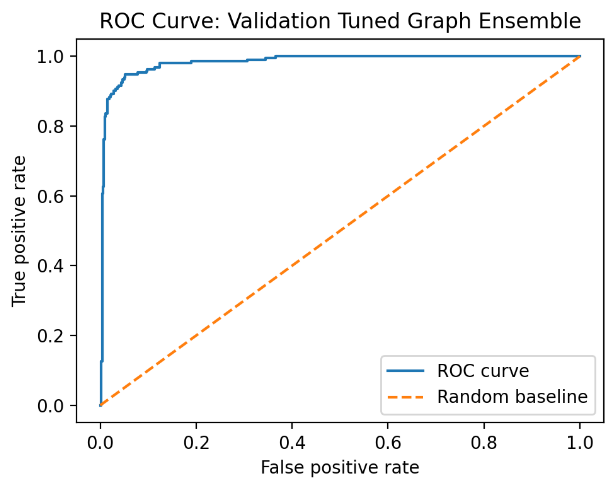

Figure 5. ROC curve of the final Validation Tuned Graph Ensemble.

5.5. Ablation Study

The ablation study compares different levels of model complexity. Traditional classifiers test the value of handcrafted EEG features. The MLP tests flat neural network learning without graph structure. Static GCN tests a fixed average graph. Dynamic GCN and Dynamic GAT test window specific graph learning. Patient Calibrated GAT and Hybrid DPC GAT test the value of patient specific graph calibration. Finally, the Validation Tuned Graph Ensemble tests whether combining dynamic and patient calibrated graph models improves the final decision.

The results show that graph based methods generally achieved higher seizure sensitivity than most traditional methods. Static GCN reached perfect recall, but its false positive rate was high. Dynamic GCN gave the strongest individual graph baseline. The final ensemble improved the balance between precision and recall, giving the best F1 score on the primary test split.

Table 4. Patient wise results for the final ensemble on unseen test patients.

Patient	Accuracy	Recall	F1	FPR
chb01	0.9519	0.9167	0.9296	0.0294
chb02	0.7500	0.9881	0.6917	0.3443
chb07	0.9345	1.0000	0.8952	0.0909

5.6. Patient Wise Evaluation

Patient wise evaluation was included to check whether the final model worked similarly across all unseen test patients. Table 4 shows the final ensemble performance for each test patient.

The patient wise results show that performance varied between patients. This is expected in cross patient EEG seizure detection because seizure patterns and normal EEG background activity can differ strongly across individuals. The model performed best on chb01 and chb07, while chb02 was more difficult and had a higher false positive rate.

5.7. Bootstrap Analysis

Bootstrap testing was used to compare the final ensemble against the strongest individual graph baseline, Dynamic GCN, on the primary test split. The bootstrap results supported the improvement trend of the final ensemble. The mean F1 improvement was 0.0161, and the final method had a higher F1 score than Dynamic GCN in 93.8 percent of bootstrap resamples. The mean false positive rate difference was negative, meaning that the ensemble usually produced fewer false positives. The final method had a lower false positive rate in 96.0 percent of bootstrap resamples.

However, the confidence interval crossed zero, so the improvement should be interpreted carefully rather than claimed as universally guaranteed. This supports the final result but also shows the need for robustness testing across different patient splits.

Table 5. Bootstrap comparison between the final ensemble and Dynamic GCN.

Metric	Mean	Lower 95	Upper 95
F1 difference	0.0161	-0.0041	0.0362
FPR difference	-0.0232	-0.0489	0.0019
Recall difference	-0.0047	-0.0149	0.0000

Table 6. Repeated patient split robustness summary.

Method	Mean F1	Mean Recall	Mean FPR
Dynamic GCN	0.6452	0.6874	0.1653
Validation Tuned Fusion	0.6311	0.6950	0.1909
Hybrid DPC GAT alpha 0.5	0.6280	0.7000	0.2009

5.8. Repeated Patient Split Robustness

To test whether the improvement was stable across different patient assignments, repeated patient split testing was performed. Three additional patient split seeds were used. For these repeated splits, Dynamic GCN, Hybrid DPC GAT, and Validation Tuned Fusion were retrained and evaluated.

The repeated split summary is shown in Table 6. Dynamic GCN achieved the highest mean F1 score across the repeated splits, while Validation Tuned Fusion achieved slightly higher mean recall than Dynamic GCN but also had a higher false positive rate. This means that the final ensemble was best on the primary split, but its improvement was not consistent across all patient splits.

This robustness result is important because it prevents overclaiming. The correct interpretation is that patient calibrated fusion improved the primary patient level test split and reduced false positives there, but the benefit is patient dependent. Cross patient EEG seizure detection remains difficult because patient variability strongly affects model generalization.

5.9. Discussion of Results

The experiments show that the proposed Validation Tuned Graph Ensemble achieved the best result on the primary patient level test split. The main practical improvement was reducing false positives compared with Dynamic GCN while keeping seizure sensitivity very high. This supports the idea that patient calibrated graph information can be useful when combined with dynamic graph models.

At the same time, the repeated split results show that the proposed method is not universally superior across all patient distributions. This is an important limitation. It suggests that patient calibration is promising, but future work should focus on making the calibration strategy more stable across larger patient groups and more varied seizure patterns.

6. Conclusion

This project presented a patient calibrated dynamic graph fusion framework for cross patient EEG seizure detection using the CHB-MIT scalp EEG dataset. The main objective was to move beyond a simple CNN or transfer learning classification task and build a more research oriented deep learning pipeline based on EEG graph modeling, patient specific calibration, validation tuned fusion, and robust evaluation.

The EEG recordings were processed into 4 second windows, and each window was represented as a dynamic brain connectivity graph. EEG channels were treated as graph nodes, while correlation based functional relationships between channels were used as graph edges. Several models were compared, including traditional machine learning methods, MLP, Static GCN, Dynamic GCN, Dynamic GAT, Patient Calibrated GAT, Hybrid DPC GAT, and the final Validation Tuned Graph Ensemble.

The main contribution of the project is the use of patient specific graph calibration. Instead of only learning global EEG connectivity patterns, the proposed method estimated normal baseline graphs for each patient using reserved non seizure calibration windows. These baseline graphs were used to construct patient calibrated graph representations. The final ensemble then combined dynamic graph models and patient calibrated graph models using validation selected weights.

On the primary patient level test split, the Validation Tuned Graph Ensemble achieved the best overall F1 score. It reached an accuracy of 0.8628, precision of 0.6840, recall sensitivity of 0.9813, specificity of 0.8142, F1 score of 0.8061, ROC AUC of 0.9836, and false positive rate of 0.1858. Compared with the strongest individual graph baseline, Dynamic GCN, the final ensemble improved F1 score and reduced false positives while maintaining very high seizure sensitivity.

However, the repeated patient split analysis showed that the improvement was not consistent across all patient assignments. Dynamic GCN achieved the strongest mean F1 score across the repeated split experiment, while the validation tuned fusion method showed slightly higher mean recall but also a higher false positive rate. This means that the proposed patient calibrated fusion method is promising, but its benefit is patient dependent.

Overall, the project satisfies the main objective by implementing a complete deep learning research pipeline with patient level splitting, multiple baselines, graph neural networks, patient calibration, ensemble fusion, bootstrap testing, repeated split robustness testing, and visual analysis. The results show that patient calibrated graph information can improve the main cross patient test result, but also confirm that cross patient EEG seizure detection remains a difficult problem due to strong patient variability.

Future work should test the method on a larger number of patients and more EEG recordings. More advanced calibration strategies could also be explored, such as adaptive thresholding per patient, temporal graph modeling, uncertainty based calibration, or domain adaptation between patients. Another useful direction would be to optimize the model for lower false alarm rates while preserving high seizure sensitivity, which is important for practical clinical use.

References

[1] Ary L. Goldberger, Luis A. N. Amaral, Leon Glass, Jeffrey M. Hausdorff, Plamen Ch. Ivanov, Roger G. Mark, Joseph E. Mietus, George B. Moody, Chung-Kang Peng, and H. Eugene Stanley. Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220, 2000.

[2] John Guttag. CHB-MIT Scalp EEG Database. *PhysioNet*, 2010. Version 1.0.0.

[3] Manhua Jia, Wenjian Liu, Junwei Duan, Long Chen, C. L. Philip Chen, Qun Wang, and Zhiguo Zhou. Efficient graph convolutional networks for seizure prediction using scalp eeg. *Frontiers in Neuroscience*, 16:967116, 2022.

[4] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.

[5] Khansa Rasheed, Adnan Qayyum, Junaid Qadir, Shobi Sivathamboo, Patrick Kwan, Levin Kuhlmann, Terence O'Brien, and Adeel Razi. Machine learning for predicting epileptic seizures using eeg signals: A review. *arXiv preprint arXiv:2002.01925*, 2020.

[6] Bahareh Salafian, Eyal Fishel Ben-Knaan, Nir Shlezinger, Sandrine de Ribaupierre, and Nariman Farsad. MICAL: Mutual information based cnn aided learned factor graphs for seizure detection from eeg signals. *arXiv preprint arXiv:2206.02298*, 2022.

[7] Ali Hossam Shoeb. *Application of Machine Learning to Epileptic Seizure Onset Detection and Treatment*. PhD thesis, Massachusetts Institute of Technology, 2009.

[8] Mohammad Khubeb Siddiqui, Ruben Morales-Menendez, Xiaodi Huang, and Nasir Hussain. A review of epileptic seizure detection using machine learning classifiers. *Brain Informatics*, 7(1):5, 2020.

[9] Petar Veličkovi'c, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Li'o, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018.